EXP NO: 14
Date: 2/10/26

Aim:

To write and implement a network packet sniffer using Raw Sockets in C to capture and parse IP, TCP and UDP network headers.

Result:-

Thus, the network packet sniffer was Sucessfully implemented using Raw Sockets in C to capture and parse IP and TCP network headers and display packet information.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/socket.h>
#include <netinet/ip.h>
#include <netinet/tcp.h>
#include <netinet/udp.h>
#include <arpa/inet.h>

#define BUFFER_SIZE 65536

void process_packet(unsigned char *buffer, int size) {
    // Extract IP Header
    struct iphdr *ip_hdr = (struct iphdr *)buffer;
    unsigned int ip_hdr_len = ip_hdr->ihl * 4;

    struct sockaddr_in source, dest;
    memset(&source, 0, sizeof(source));
    source.sin_addr.s_addr = ip_hdr->saddr;

    memset(&dest, 0, sizeof(dest));
    dest.sin_addr.s_addr = ip_hdr->daddr;

    printf("\n================ Packet Captured
================\n");
```